ABHILASH K. CHANDEL

Assistant Professor and Extension Specialist
Precision Agriculture Technologies and Data Management
Department of Biological Systems Engineering
Tidewater Agricultural Research and Extension Center
6321, Holland Rd. Suffolk, VA 23437
Email: abhilash.iitkharagpur@gmail.com; Phone: +1-509-592-1389
Webpage: https://www.arec.vaes.vt.edu/arec/tidewater/people/faculty/abhilash-chandel.html
Google Scholar: https://scholar.google.com/citations?user=xCaMSKYAAAAJ&hl=en&oi=ao
LinkedIn: https://www.linkedin.com/in/abhilash-kumar-chandel-ph-d-2bba5227/

Program Focus: Research (70%) & Extension (30%)

My research and extension programs are focused on design, development, and deployment of "Sensing, automation, and data-driven solutions for spatiotemporal and targeted management of agricultural production inputs for field and specialty crops". Specifically, my programs utilize

- Ag-sensing: ground (fixed and mobile), smartphone, aerial, and satellite-based
- Internet-of-things enabled cloud and edge computing/actuation systems
- Computer vision, AI, and big-data analytics
- Precision crop input application systems
- Agricultural machinery systems: autonomous and semi-autonomous
- Decision support tools (web and smartphone-based)
- Crop biophysical sensing and modeling

I. Executive Summary

Education

Ph.D., May 2021	Biological and Agricultural Engineering, Washington State University, WA.
M.Tech., May 2015	Farm Machinery and Power Engineering, Indian Institute of Technology (IIT) Kharagpur, India.
B.Tech., May 2015	Agricultural and Food Engineering, Indian Institute of Technology (IIT) Kharagpur, India.

Professional appointments

Jan 2022 – present	Assistant Professor and Extension Specialist, Biological Systems Engineering, Virginia Tech
Jun 2021 – Dec 2021	Postdoctoral Research Associate, Biological Systems Engineering, Washington State University
Jul 2015 – Feb 2017	Junior Project Officer, Indian Institute of Technology Kharagpur, India

Selected publications since joining Virginia Tech (Lead author bolded; advisees under my direct mentorship are indicated – [P]postdoc advisee, [G]graduate student advisee; [C]corresponding author)

Selected Peer-reviewed journal publications

Chandel, N.S., Chakraborty, S.K., Chandel, A.K.[C], Upadhyay, A., Parmar, B.S., Manjhi, M., Jat, D., 2026. Time-of-Flight (ToF) camera technology for high throughput holistic phenotyping and canopy volume measurement of horticultural crops. *Scientia Horticulturae*, *358*, p.114742. https://doi.org/10.1016/j.scienta.2026.114742

Sahayaraj, S.R.E.[G], Chandel, A.K.[C], Jjagwe, P.[G], Balota, M., Forehand, J., Chappell, M., 2026. Agroclimatic Interactions Influencing Maturity of Virginia-Type Peanuts in Humid Subtropical Environments: Toward Sustainable Peanut Production. *Agricultural Environment and Sustainability*, p.100025. https://doi.org/10.1016/j.ages.2026.100025

Sahayaraj, S.R.E.[G], Chandel, A.K.[C], Jjagwe, P.[G], Vennam, R.R., Balota, M., Manimozhian, A.[P], 2026. Toward Advanced Sensing and Data-Driven Approaches for Maturity Assessment of Indeterminate Peanut Cropping Systems: Review of Current State and Prospects. *Sensors*, *26*(7), p.2208. https://doi.org/10.3390/s26072208

Manimozhian, A.[P], Chandel, A.K.[C], 2026. Thermal Infrared Technologies for Precision Agriculture: A ROSES-Guided Systematic Evidence Synthesis on Platforms, Calibration, and Digital Analytics. *Agricultural Environment and Sustainability*, p.100014. https://doi.org/10.1016/j.ages.2026.100014

Sarr, A.[G], Chandel, A.K.[C], Diop, L., Soro, Y.M., Tossa, A.K., Hota, S., Manimozhian, A., 2026. Agroclimatic Sensing, Communication, and Computational Systems-Based Methods and Technologies for Precision Irrigation Management: Current State and Prospects. *Computers*, 15(2), p.137. https://doi.org/10.3390/computers15020137

Vennam, R.R., Chandel, A.K., Haak, D.C., Oakes, J., Balota, M.[C], 2026. Leaf wilting as a phenotypic indicator of heat and drought stress in crops: an overview of physiological mechanisms and machine learning applications. *Discover Agriculture*, 4, p.32. https://doi.org/10.1007/s44279-026-00506-6

Nkwocha, C.L.[G], Chandel, A.K[C]., 2025. Towards an End-to-End Digital Framework for Precision Crop Disease Diagnosis and Management Based on Emerging Sensing and Computing Technologies: State over Past Decade and Prospects. *Computers*, 14(10), p.443. https://doi.org/10.3390/computers14100443

Chouriya, A., Soni, P., Chandel, A.K.[C], Patel, A.K., 2025. An AI-Enabled System for Automated Plant Detection and Site-Specific Fertilizer Application for Cotton Crops. *Automation*, 6(4), p.53. https://doi.org/10.3390/automation6040053

Chandel, A.K.[C]**,** Khot, L.R.[C], Stöckle, C.O., Kalcsits, L., Mantle, S., Rathnayake, A.P. and Peters, R.T. 2025. Canopy transpiration mapping in an apple orchard using high-resolution airborne spectral and thermal imagery with weather data. *Agriengineering*, 7(5), p.154. https://doi.org/10.3390/agriengineering7050154

Chandel, N.S., Chakraborty, S.K., Chandel, A.K.[C], Dubey, K., Subeesh, A., Jat, D., Rajwade, Y., 2024. State-of-the-art AI-enabled mobile device for real-time water stress detection of field crops. *Engineering Applications of Artificial Intelligence*, 131, 107863. https://doi.org/10.1016/j.engappai.2024.107863

Jjagwe, P.[G], Chandel, A.K.[C], Langston, D.B., 2024. Impact Assessment of Nematode Infestation on Soybean Crop Production Using Aerial Multispectral Imagery and Machine Learning. *Applied Sciences*, 14(13), p.5482. https://doi.org/10.3390/app14135482

Jjagwe, P.[G], Chandel, A.K. [C], Langston, D., 2023. Pre-harvest corn grain moisture estimation using aerial multispectral imagery and machine learning techniques. *Land*, 12(12), p.2188. https://doi.org/10.3390/land12122188

Kumar, S.P., Tewari, V.K., Chandel, A.K.[C], Mehta, C.R., Pareek, C.M., Chethan, C.R., Nare, B., 2023. Modelling specific energy requirement for a power-operated vertical axis rotor type intra-row weeding tool using artificial neural network. *Applied Sciences*, 13(18), p.10084. https://doi.org/10.3390/app131810084

Modi, R., Chandel, A.K. [C], Chandel, N.S., Dubey, K., Subeesh, A., Singh, A., Jat, D., Kancheti, M., 2023. State-of-art computer vision techniques for automated sugarcane lodging classification. *Field Crops Research*, 291, p.108797. https://doi.org/10.1016/j.fcr.2022.108797

Chandel, A.K.[C], Moyer, M.M., Keller, M., Khot, L.R., Hoheisel, G.A., 2022. Soil and climate geographic information system data-derived risk mapping for grape Phylloxera in Washington state. *Frontiers in Plant Science*, 13, pp.827393-827393. https://doi.org/10.3389/fpls.2022.827393

Chandel, N.S., Rajwade, Y.A., Dubey, K., Chandel, A.K.[C], Subeesh, A., Tiwari, M.K., 2022. Water stress identification of winter wheat crop with state-of-the-art ai techniques and high-resolution thermal-RGB imagery. *Plants*, 11(23), p.3344. https://doi.org/10.3390/plants11233344

Numbered Extension publications

Manimozhian, A.[P], Jjagwe, P.[G], Chandel, A.K.[C], 2026. Mapping Cotton Boll Opening at Field-Scale (%Open Bolls) Using UAV Imagery. VCE Publications, BSE-385NP.

Jjagwe, P.[G], Flessner, M., Holshouser, D., Chandel, A.K.[C], Hota, S., 2025. Potential of Controlled Traffic Farming for Enhanced Soil Health and Productivity. VCE Publications, BSE-374NP.

Chandel, A.K.[C]**,** Konduru, L.V., Kanakamedala, V.C., Maurya A.[P], Hota, S. 2025. Agroclimate Viewer & Planner App. VCE Publications, BSE-372NP.

Chandel, A.K.[C]**,** Konduru, L.V., Kanakamedala, V.C., Maurya A.[P], Hota, S. 2025. Agroclimate Viewer & Planner App (NDVI only). VCE Publications, BSE-371NP.

Chandel, A.K.[C]**,** Jjagwe, P.[G], Langston, D., 2024. Drone imaging to evaluate impact of Nematodes on Soybean Yield. VCE Publications, BSE-362NP.

Balota, M., Chandel, A.K., 2024. Prohexadione Calcium or Seed Aging?. VCE Publications, SPES-581NP.

Raman, R., Balota, M.[C], Chandel, A.K., Jjagwe, P.[G], 2024. Faba Bean: A Multipurpose Specialty Crop for the Mid-Atlantic USA. VCE Publications, SPES-590NP.

Chandel, A.K.[C], Langston, D., 2023. Aerial multispectral imagery for high-throughput mapping of spatial corn yield potentials. VCE Publications, SPES-526NP.

Chandel, A.K.[C], Langston, D., 2023. Aerial multispectral imagery for high-throughput mapping of spatial soybean yield potentials. VCE Publications, SPES-527NP.

Chandel A.K.[C], 2023. Aerial imagery to improve disease diagnosis and management in field crops. VCE Publications, SPES-515NP.

Selected service since joining Virginia Tech

Conference Chair, Autonomous Air and Ground Sensing Systems for Agricultural Optimization and Phenotyping XI. SPIE, The International Society for Optics and Photonics. November 2025-present

Member, ASABE EOPD-208 Extension committee. November 2025-present.

Member, International Organization for Standardization Technical Committee 347 on Data-Driven Agrifood Systems. October 2025-present.

Journal Associate Editor, Irrigation Science and Frontiers in Plant Science. October 2025-present.

Chair, USDA multistate group S1098: Autonomy for Agricultural Production, Processing, and Research to Advance Food Security through Sustainable and Climate-Smart Methods. Chair, April 2025-present.

Member, Virginia Tech Tidewater Agricultural Research and Extension Center. Faculty search committee. January 2023-Oct 2024.

Member, USDA open-data-framework. National Agricultural Producer Data Cooperative. Develop a national cyberinfrastructure to drive innovation and increase production in agriculture. August 2022-August 2025.

Member, Virginia Tech Department of Biological Systems Engineering. Awards and honorifics committee. June 2022-present.
Member, Virginia Tech Department of Biological Systems Engineering. Extension committee. June 2022-present.

II. Teaching and Advising Effectiveness

A. Chronological list and/or table of non-credit courses, workshops, and other related outreach and/or extension teaching since the date of appointment to Virginia Tech (GGraduate Advisee. Presenter is bolded)

Summary of Extension Programs, Seminars, and Workshops

Subject Area	Number of programs	Contact time (~h)	Attendees
Digital user tools for agriculture	15	736	3,577
Integrative precision agriculture	23*	39	1,133
Unoccupied Aerial Systems (UASs) for agriculture	12*	118	936
Sensors and data in agriculture	6	47	325
Total	**56**	**940**	**5,971**

*event listed under VI.D.4

<u>Extension presentations/talks</u> (44)

<u>Digital user tools for agriculture</u> (15)

Chandel, A.K. A free web tool for agroclimatic assessment of crop fields. February 25, 2025–May 31, 2026. (Webtool online visits: ~2,900, Avg. time per visit: ~16 min, Total contact time: ~728 h).
Chandel, A.K. AgroVAP for precision berry production. June 3, 2026. Virginia Beach, VA. (Contact time: ~25 min, Attendees: ~40).
Chandel, A.K. AgroVAP for precision row crop production. April 22, 2026. Suffolk, VA. (Contact time: ~30 min, Attendees: ~25).
Chandel, A.K. AgroVAP: A farmer friendly tool for field level precision agricultural management. February 13, 2026. Chesapeake, VA. (Contact time: ~25 min, Attendees: ~70).
Chandel, A.K. Agroclimate Viewer and Planner App (AgroVAP) for Soybean Growers. Soybean field day. September 11, 2025. Warsaw, VA. (Contact time: ~25 min, Attendees: ~70).
Chandel, A.K. Introduction to Agroclimate viewer and planner app (AgroVAP) for precision agriculture. APRES field tour. July 11, 2025. Goodrich farms, VA. (Contact time: ~20 min, Attendees: ~70).
Chandel, A.K. Agroclimate viewer and planner app: A free platform for crop management decision making. Berry field day. June 4, 2025. Virginia Beach, VA. (Contact time: ~15 min, Attendees: ~40).
Chandel, A.K. Know how your crop is doing with Agroclimate Viewer and Planner App (Features and Utilization). Technology, Economy, and Wellness workshop for farmers and Extension agents. April 9, 2025. Suffolk, VA. (Contact time: ~1 h, Attendees: ~65).
Chandel, A.K. Agroclimate viewer and planner app. Cover crop field day. March 21, 2025. Suffolk, VA. (Contact time: ~15 min, Attendees: ~100).

Chandel, A.K. Webtool for producers to monitor field-level agroclimate for precision agriculture operations. Virginia Farm Bureau Southeastern Producers Meeting. March 10, 2025. Smithfield, VA. (Contact time: ~30 min, Attendees: ~50).
Chandel, A.K. A free web tool for agroclimatic assessment of crop fields. Strawberry School and Tradeshow. February 25, 2025. Virginia Beach, VA. (Contact time: ~30 min, Attendees: ~80).
Chandel, A.K. A free web tool for agroclimatic assessment of crop fields. HR Fruit and Vegetable Workshop. February 19, 2025. Chesapeake, VA. (Contact time: ~30 min, Attendees: ~45).
Chandel, A.K. A free web tool for agroclimatic assessment of crop fields. Precision Agriculture for Farmers. February 12, 2025. Randolph farms, Virginia State University, Petersburg, VA. (Contact time: ~1 h, Attendees: ~10).
Chandel, A.K. A free web tool for agroclimatic assessment of crop fields. January 31, 2025. Virtual. (Contact time: ~1 h, Attendees: ~10).
Chandel, A.K. A free web tool for agroclimatic assessment of crop fields. January 16, 2025. Tidewater AREC. (Contact time: ~45 min, Attendees: 2).

Integrative precision agriculture (14)
Chandel, A.K. AI for Agriculture: from basics to practical applications. March 27, 2026. Online webinar hosted by Louisiana State University. (Contact time: ~60 min, Attendees: ~20).
Chandel, A.K. Jjagwe, P[G], Chijioke N[G]. Introduction to Precision Ag for Ag Technology Students. October 09, 2025. Suffolk, VA. (Contact time: ~20 min, Attendees: ~50).
Chandel, A.K. Jjagwe, P[G], Chijioke N[G]. Precision Ag technologies for agriculture and natural resources. Row Crop Training, September 25, 2025. Suffolk, VA. (Contact time: ~20 min, Attendees: ~30).
Chandel, A.K.. Precision Ag updates for peanut production. Peanut variety and quality evaluation field day. September 17, 2025. Williamston, NC. (Contact time: ~20 min, Attendees: ~50).
Chandel, A.K. Raymond, S[G], Jjagwe, P[G]. Technologies for peanut production management. Peanut variety and quality evaluation field day. September 11, 2024. Williamston, NC. (Contact time: ~30 min, Attendees: ~50).
Chandel, A.K. Raymond, S[G], Jjagwe, P[G]. Precision agriculture technologies to support peanut production. Cotton and Peanut field day. August 18, 2024. Tidewater AREC, Suffolk, VA. (Contact time: ~30 min, Attendees: ~60).
Chandel, A.K. Raymond, S.[G], Jjagwe, P[G]. Precision agriculture technologies to support efficient crop production in VA. Virginia Ag Expo. August 1, 2024, Champlain, VA. (Contact time: ~30 min, Attendees: ~30).
Chandel, A.K. Precision agriculture technologies to support efficient crop production in VA. The Berry field day. June 4, 2024. Hampton Roads AREC, Virginia Beach, VA. (Contact time: ~30 min, Attendees: ~40).
Wolfe, M., Xia, K., Chandel, A.K. Precision vegetable production using technology. Southern Piedmont AREC Field Day. July 27, 2023. Southern Piedmont AREC, Blackstone, VA. (Contact time: ~40 min, Attendees: ~100).
Wolfe, M., Xia, K., Chandel, A.K. Precision Ag for vegetables seminar and focus group meeting. April 13, 2023. Randolph farms, Virginia State University, Petersburg, VA. (Contact time: ~1 h, Attendees: ~50).
Chandel, A.K. Innovative Precision Agriculture “VCE Growers meeting”. September 28, 2022. McKenney, VA. (Contact time: ~20 min, Attendees: ~20).

Chandel, A.K., Posadas, B. Crop production management with technology survey. Small grains field day. May 19, 2022. Eastern Virginia AREC, Warsaw, VA. (Contact time: ~40 min, Attendees: ~75).

Chandel, A.K. Introduction to precision agriculture. Virginia peanut grower annual meeting, Feb 24, 2022. Franklin, VA. (Contact time: ~10 min. Attendees: ~150).

Chandel, A.K. Introduction to precision agriculture in regional crop production management. Feb 24, 2022. Virginia Northern Neck Crops Conference. (Contact time: ~15 min. Attendees: ~150).

Unoccupied Aerial Systems (UASs) for agriculture (10)

Chandel, A.K. Drone applications in strawberry production. Mid-Atlantic strawberry field day. February 26, 2024. Chesapeake, VA. (Contact time: ~1 h, Attendees: ~60).

Chandel, A.K., Raymond, S[G]. Drones and AI for peanut maturity mapping, Virginia peanut grower annual meeting, Franklin, VA. Feb 22, 2024. (Contact time: ~10 min. Attendees: ~150).

Chandel, A.K. Drone applications in agriculture. Private applicator recertification training. December 14, 2023. Rapidan, VA (Virtual). (Contact time: ~45 min, Attendees: ~100).

Chandel, A.K. Jjagwe, P[G]**.** Estimating optimum peanut maturity using technology. Virginia agricultural technology student engagement. October 19, 2023. Tidewater AREC, Suffolk, VA. (Contact time: ~30 min, Attendees: ~70).

Chandel, A.K. High-throughput estimation of optimum peanut maturity using technology. Peanut variety and quality evaluation field day. September 13, 2023. Williamston, NC. (Contact time: ~30 min, Attendees: ~50).

Chandel, A.K. UAVs in agriculture: current status, challenges, and future roadmaps. Virginia Crop Production Association Conference. January 18, 2023. Richmond, VA. (Contact time: ~50 min, Attendees: ~250).

Chandel, A.K. Drones for mapping and spraying. NSV Commercial Pesticide Meeting. January 10, 2023. Winchester, VA. (Contact time: ~45 min, Attendees: ~130).

Chandel, A.K. Peanut Soil & Water Conservation District Specialists training on drone imagery processing, Sep 16, 2022. (Contact time: ~1 h, Attendees: 1).

Balota, M., Chandel, A.K. Peanut pod blasting and aerial imaging campaigns. September 12, 2022. Windsor, VA. (Contact time: ~3 h, Attendees: ~65).

Sensors and data in agriculture (5)

Chandel, A.K., Balota, M. Machine learning analytics for crop trait prediction. Training for students of Uganda, Ghana, and Senegal. June 16– August 10, 2023. Virtual training. (Contact time: ~32 h, Attendees: ~5).

Chandel, A.K. Utilizing satellite imagery and data analytics to realize problems in soybean and corn growing fields. Central Virginia Crops Expo. August 26, 2022. Brookneal, VA. (Contact time: ~30 min, Attendees: ~30).

Chandel, A.K. Demonstration of drones, sensors, and data for production management. Cotton field day. August 18, 2022. Tidewater AREC, Suffolk, VA. (Contact time: ~30 min, Attendees: ~60).

Chandel, A.K. Accessing satellite imagery and analytics to realize problems in soybean and corn growing fields. Virginia Agricultural Expo. August 4, 2022. Port Royal, VA. (Contact time: ~5 h, Attendees: ~200).

Chandel, A.K. Utilizing remote sensing and precision agriculture for understanding the impacts of control traffic farming. March 7, 2022. Shepherd Grain Farms, Blackstone, VA. (Contact time: ~3 h, Attendees: ~10).

Focus group meetings organized (5)

Integrative precision agriculture (5)

Chandel, A.K., McCall, D., Oakes, J. Virginia peanut production with state-of-the-art technology, focus group meeting. March 8, 2024. Tidewater AREC, Suffolk, VA. (Contact time: ~3 h, Attendees: ~20).

Chappell, M., Langston D., Chandel, A.K., Frame, H., Balota, M. Melone, S. Virginia peanut producers focus group meeting. July 27, 2023. Tidewater AREC, Suffolk, VA. (Contact time: ~2 h, Attendees: ~20).

Parraga-Estrada, K., Chandel, A.K. Precision aquaculture online focus group meeting. April 14, 2023. (Contact time: ~1 h, Attendees: 2).

Chandel, A.K. Field crop producer focus group meeting. January 19, 2023. Tidewater AREC, Suffolk, VA. (Contact time: ~3 h, Attendees: ~30).

Chandel, A.K. Planning meeting for VA Ag Expo, Mill Creek farms. July 10, 2022. Camden, VA. (Contact time: ~2 h, Attendees: ~25).

Workshops organized (5)

Integrative precision agriculture (3)

Chandel, A.K. Precision Ag workshop 2026 for VCE agents and farmers. April 22, 2026. Tidewater AREC, Suffolk, VA. (Contact time: ~6 h, Attendees: ~25).

Chandel, A.K. Technology, economy, and wellness workshop 2025 for VCE agents and farmers. April 9, 2025. Tidewater AREC, Suffolk, VA. (Contact time: ~5 h, Attendees: ~65).

Chandel, A.K. Precision Ag Technology Expo 2024. June 28, 2024. Tidewater AREC, Suffolk, VA. (Contact time: ~7 h, Attendees: ~50).

Unoccupied Aerial Systems (UASs) for agriculture (2)

Chandel, A.K. Drone school 2024. For VCE agents and students of Uganda, Ghana, Senegal, Malawi, and Burkina Faso. March 18–April 30, 2024. Tidewater AREC, Suffolk, VA. (Contact time: ~100 h, Attendees: ~15).

Chandel, A.K. Drone data analytics workshop. April 19, 2024. Tidewater AREC, Suffolk, VA. (Contact time: ~8 h, Attendees: ~25).

Sensors and data in agriculture (1)

Chandel, A.K. Workshop on Crop water use mapping using ET sensors for precision irrigation. June 25, 2024. Tidewater AREC, Suffolk, VA. (Contact time: ~6 h, Attendees: ~20).

III. Research and Creative Activities

A. Awards, prizes, and recognitions

Since joining VT (1)

2023 Outstanding reviewer award, ASABE

Prior to joining VT (11)

2021 Finalist, WSU plant science and Corteva symposium
2021 Graduate Student Leadership Award 2021, CAHNRS, WSU
2021 ASABE Agricultural Equipment Technology Conference Participation award

2020 Microsoft Artificial Intelligence for Earth
2020 Walter and Vinnie Hinz Outstanding Graduate Student, BSE, WSU
2020 Winner, Digital Ag-Athon 2020, jointly organized by Microsoft and WSU
2020 BSE, WSU. Virtual Conference Participation Award
2019 Water Smart Innovation Conference Travel Award
2019 Alaska Airlines Travel Award
2019 ASABE AIM Travel Award, BSE, WSU
2010 Merit-cum-Means (MCM) undergraduate scholarship

B. List of publications and creative scholarship (Lead author bolded; advisees under my direct mentorship are indicated – [P]postdoc advisee, [G]graduate student advisee; [C]corresponding author; all citations values taken from Google Scholar in March 2026; 5-year impact factor for journals are reported from Web of Science in March 2026)

	Since joining Virginia Tech			**Prior to joining Virginia Tech**		
Type of Publication	Student /Postdoc Lead Author[1]	Lead or Corr. Author	Co-Author	Lead or Corr. Author	Co-Author	Total
Book Chapters	1	2	–	–	1	3
Papers in Refereed Journals	8	17	12	9	12	50
Refereed Conference Proceedings	5	6	1	3	3	13
Numbered (peer-reviewed) Extension Publications	2	8	2	–	–	10
Papers and Posters Presented at Professional Meetings/ Abstracts	18	13	49	12	13	87
Other Papers and Reports	–	–	4	–	3	7
Other Outreach Publications (VI.D.3)	–	5	2	1	2	10
Total	**34**	**50**	**70**	**25**	**34**	**180**

[1]Not included in row totals. Google Scholar: citations: 1062, h-index: 18, i10-Index: 32. (Accessed May 2026)

1. **Book chapters** (3 total, 2 since joining VT)

Since joining VT (2)

Manimozhian, A.[P], Chandel, A.K.[C], 2026. Thermal Infrared Sensing for Precision Agriculture: Fundamentals and Applications. Book Chapter in Taylor and Francis CRC press (Accepted).

Chandel, A.K.[C]**,** 2024. Satellite-based remote sensing approaches for estimating evapotranspiration from agricultural systems. In Digital Agriculture: *A Solution for Sustainable Food and Nutritional Security* (pp. 281-323). Springer, New York, NY. ISSN 2731-3506. (Citations = 1)

Prior to joining VT (1)

Sinha, R., Khot, L.R.[C], Gao, Z., Chandel, A.K., 2021. Sensors Part III: spectral sensing and data analysis. In Fundamentals of agricultural and field robotics, *Agriculture Automation and Control* (pp.79-110). Springer, New York, NY. ISSN 2731-3506. (Citations = 3)

My role: Conceptualization, writing – original draft, review and editing.

2. Papers in refereed journals (both print and electronic, 50 total, 29 since joining VT)

Since joining VT (29)

Chandel, N.S., Chakraborty, S.K., Chandel, A.K.[C], Upadhyay, A., Parmar, B.S., Manjhi, M., Jat, D., 2026. Time-of-Flight (ToF) camera technology for high throughput holistic phenotyping and canopy volume measurement of horticultural crops. *Scientia Horticulturae*, *358*, p.114742. https://doi.org/10.1016/j.scienta.2026.114742

Sahayaraj, S.R.E.[G], Chandel, A.K.[C], Jjagwe, P.[G], Balota, M., Forehand, J., Chappell, M., 2026. Agroclimatic Interactions Influencing Maturity of Virginia-Type Peanuts in Humid Subtropical Environments: Toward Sustainable Peanut Production. *Agricultural Environment and Sustainability*, p.100025. https://doi.org/10.1016/j.ages.2026.100025

Sahayaraj, S.R.E.[G], Chandel, A.K.[C], Jjagwe, P.[G], Vennam, R.R., Balota, M., Manimozhian, A.[P], 2026. Toward Advanced Sensing and Data-Driven Approaches for Maturity Assessment of Indeterminate Peanut Cropping Systems: Review of Current State and Prospects. *Sensors*, *26*(7), p.2208. https://doi.org/10.3390/s26072208

Manimozhian, A.[P], Chandel, A.K.[C], 2026. Thermal Infrared Technologies for Precision Agriculture: A ROSES-Guided Systematic Evidence Synthesis on Platforms, Calibration, and Digital Analytics. *Agricultural Environment and Sustainability*, p.100014. https://doi.org/10.1016/j.ages.2026.100014

Sarr, A.[G], Chandel, A.K.[C], Diop, L., Soro, Y.M., Tossa, A.K., Hota, S., Manimozhian, A. 2026. Agroclimatic Sensing, Communication, and Computational Systems-Based Methods and Technologies for Precision Irrigation Management: Current State and Prospects. *Computers*, 15(2), p.137. https://doi.org/10.3390/computers15020137

Vennam, R.R., Chandel, A.K., Haak, D.C., Oakes, J., Balota, M.[C], 2026. Leaf wilting as a phenotypic indicator of heat and drought stress in crops: an overview of physiological mechanisms and machine learning applications. *Discover Agriculture*, 4, p.32. https://doi.org/10.1007/s44279-026-00506-6

Nkwocha, C.L.[G], Chandel, A.K[C]., 2025. Towards an End-to-End Digital Framework for Precision Crop Disease Diagnosis and Management Based on Emerging Sensing and Computing Technologies: State over Past Decade and Prospects. *Computers*, 14(10), p.443. https://doi.org/10.3390/computers14100443

Chouriya, A., Soni, P., Chandel, A.K.[C], and Patel, A.K. 2025. An AI-Enabled System for Automated Plant Detection and Site-Specific Fertilizer Application for Cotton Crops. *Automation*, 6(4), p.53. https://doi.org/10.3390/automation6040053

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3. **Papers in refereed conference proceedings** (13 total, 7 since joining VT)

Since joining VT (7)

Jjagwe, P.[G]**,** Chandel A.K.[C], Balota, M., Seetharaman, R.[G], Nkwocha, C.[G], Bandi, S., Konduru, L.V., Yadav, V., 2026. Weed detection in faba bean using unoccupied aerial system imagery and computer vision models. In: Autonomous Air and Ground Sensing Systems for Agricultural Optimization and Phenotyping XI. *SPIE*. Paper no. 14045-17.

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Raymond, S.[G]**,** Chandel A.K.[C], Balota, M., Chappell, M., Sridhar, V., 2025. Leveraging Stacked Generalization for Peanut Maturity Mapping Using Aerial Multispectral Imagery and Growing Degree Days. In: Autonomous Air and Ground Sensing Systems for Agricultural Optimization and Phenotyping X. *SPIE.* 13475, pp. 202-212. https://doi.org/10.1117/12.3054125

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Tolomelli, G.[C]**,** Kothawade, G.S., Chandel, A.K., Manfrini, L., Jacoby, P., Khot, L.R., 2022, November. Aerial-RGB imagery-based 3D canopy reconstruction and mapping of grapevines for precision management. In: 2022 IEEE Workshop on Metrology for Agriculture and Forestry, *IEEE Xplore*, pp.203-207. https://doi.org/10.1109/MetroAgriFor55389.2022.9965062

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Molaei, B.[C], Peters, R.T., Chandel, A.K., Khot, L.R., Stöckle, C.O., Campbell, C.S., 2021. Comparing downwind evapotranspiration suppression in LESA and MESA irrigation systems. In: 6th Decennial National Irrigation Symposium, 6-8, December 2021, San Diego, California (p. 1). *American Society of Agricultural and Biological Engineers*. https://doi.org/10.13031/irrig.2020-138

Chandel, A.K.[C], Khot, L.R., Brown, D.J., Peters, T.R., Stöckle, C.O., Mantle, S., 2021. Geospatial apple canopy transpiration mapping: effect of in-field and open-field weather. In: 2021 IEEE International Workshop on Metrology for Agriculture and Forestry, *IEEE Xplore*. pp.182-186. https://doi.org/10.1109/MetroAgriFor52389.2021.9628856

Kothawade, G.S.[C], Chandel, A.K., Schrader, M.J., Rathnayake, A.P., Khot, L.R., 2021. High throughput canopy characterization of a commercial apple orchard using a small UAS equipped with a consumer-grade camera. In: 2021 IEEE International Workshop on Metrology for Agriculture and Forestry, *IEEE Xplore*. pp.177-181. https://doi.org/10.1109/MetroAgriFor52389.2021.9628564

Chandel, A.K.[C], Khot, L.R., Sallato, B.C., 2020. Towards rapid detection and mapping of powdery mildew in apple orchards. In: 2020 IEEE International Workshop on Metrology for Agriculture and Forestry, *IEEE Xplore*. pp.288-292. https://doi.org/10.1109/10.1109/MetroAgriFor50201.2020.9277544

Chandel, A.K.[C], Khot, L.R., Stöckle, C.O., Peters R.T., Mantle, S., 2020. Spatiotemporal water use mapping of a commercial apple orchard using UAS based spectral imagery. In: 2020 IEEE International Workshop on Metrology for Agriculture and Forestry, *IEEE Xplore*. pp.268-272. https://doi.org/10.1109/MetroAgriFor50201.2020.9277550

Amogi. B.R.[C]**,** Chandel, A.K., Khot, L.R., Jacoby, P.W., 2020. A mobile thermal-RGB imaging tool for mapping crop water stress of grapevines. 2020 IEEE International Workshop on Metrology for Agriculture and Forestry, *IEEE Xplore*. pp.293-297. https://doi.org/293-297.10.1109/MetroAgriFor50201.2020.9277545

4. **Numbered extension publications** (10 total, 10 since joining VT)

Since joining VT (10)

Manimozhian, A.P, Jjagwe, P.G, Chandel, A.K.C, 2026. Mapping Cotton Boll Opening at Field-Scale (%Open Bolls) Using UAV Imagery. VCE Publications, BSE-385NP.

Jjagwe, P.G, Flessner, M., Holshouser, D., Chandel, A.K.C, Hota, S., 2025. Potential of Controlled Traffic Farming for Enhanced Soil Health and Productivity. VCE Publications, BSE-374NP.

Chandel, A.K.C**,** Konduru, L.V., Kanakamedala, V.C., Maurya A.P, Hota, S. 2025. Agroclimate Viewer & Planner App. VCE Publications, BSE-372NP.

Chandel, A.K.C**,** Konduru, L.V., Kanakamedala, V.C., Maurya A.P, Hota, S. 2025. Agroclimate Viewer & Planner App (NDVI only). VCE Publications, BSE-371NP.

Chandel, A.K.C**,** Jjagwe, P.G, Langston, D., 2024. Drone imaging to evaluate impact of Nematodes on Soybean Yield. VCE Publications, BSE-362NP.

Raman, R.C, Balota M., Chandel, A.K., Jjagwe, P.G, 2024. Faba Bean: A Multipurpose Specialty Crop for the Mid-Atlantic USA. VCE Publications, SPES-590NP.

Balota M.C, Chandel, A.K., 2024. Prohexadione Calcium or seed aging? VCE Publications, SPES-581NP.

Chandel, A.K.C, Langston, D., 2023. Aerial multispectral imagery for high-throughput mapping of spatial corn yield potentials. VCE Publications, SPES-526NP.

Chandel, A.K.C, Langston, D., 2023. Aerial multispectral imagery for high-throughput mapping of spatial soybean yield potentials. VCE Publications, SPES-527NP.

Chandel A.K.C, 2023. Aerial imagery to improve disease diagnosis and management in field crops. VCE Publications, SPES-515NP.

5. **Papers and posters presented at professional meetings** (87 total, 62 since joining VT)

Since joining VT (62)

Jjagwe, P.G**,** Chandel A.K., Balota, M., Seetharaman, R.G, Nkwocha, C.G, Bandi, S., Konduru, L.V., Yadav, V., 2025. Weed detection in faba bean using unoccupied aerial system imagery and computer vision models. SPIE Defense + Commercial Sensing exhibition, April 26-30, 2026, National Harbor, MD. (oral presentation).

Nkwocha, C.G**,** Chandel A.K., 2026. Development of a low-cost UGV for autonomous crop monitoring: navigation optimization using boustrophedon path planning and pure pursuit tracking algorithms. SPIE Defense + Commercial Sensing exhibition, April 26-30, 2026, National Harbor, MD. (oral presentation).

Jjagwe, P.G, Chandel, A.K., Balota, M., Seetharaman, R.G, Raman, R., 2026. Faba bean crop identification and mapping from UAS RGB imagery using deep learning approaches. AI in Agriculture Conference, March 31- April 2, 2026, Raleigh, NC. (poster presentation).

Nkwocha, C.G, Chandel, A.K., Langston, D.B., Batarseh, F.A., Samtani, J.B., 2026. UGV-based sensing and AI-edge computing for an end-to-end autonomous crop disease diagnosis and management. AI in Agriculture Conference, March 31- April 2, 2026, Raleigh, NC. (poster presentation).

Raymond, S.G, Chandel, A.K., Balota, M., 2026. Web-Based Platform for Assessing Peanut Pod Maturity Using Aerial Spectral Imagery and Multiview

Machine Learning. AI in Agriculture Conference, March 31- April 2, 2026, Raleigh, NC. (poster presentation).

Tate, B., Frame, W., Chandel, A.K., Eick, M., Guo, F., Shortridge, J., Stewart, R. (2025). Nitrogen Prediction Algorithm for Cotton Grown in the Upper Southeast Coastal Plain. In 2026 Beltwide Cotton Conference. January 14-16, 2026. New Orleans, LA. (oral presentation).

Zheng, Y., Jjagwe, P.[G], Granja, M., Chandel, A.K., Ortel, C., Zhang, B. (2025). Multimodal UAS-Based High-Throughput Phenotyping and Machine Learning for Early-Season Yield Prediction in Soybean Breeding. Translational Plant Sciences Center Symposium. February 21, 2025. Blacksburg, VA. (oral presentation).

Zheng, Y., Jjagwe, P.[G], Granja, M., Chandel, A.K, Ortel, C., Zhang, B., 2025. Multimodal UAS-Based High-Throughput Phenotyping and Machine Learning for Early-Season Yield Prediction in Soybean Breeding. In Center for Advanced Innovation in Agriculture (CAIA) 2025 Big Event. May 6, 2025. Blacksburg, VA. (poster presentation).

Zheng, Y., Jjagwe, P.[G], Ogando do Granja, M., Chandel, A.K., Ortel, C., Zhang, B., 2025. Integrating UAS-Based Phenotyping and Machine Learning for Yield Prediction in Soybean. Joint MAS-ASPB (Mid-Atlantic Section of the American Society of Plant Biologists) and UMD (University of Maryland) Plant Symposium. May 28-29, 2025. College Park, MD. (oral presentation).

Chandel, A.K., 2025. Free Webtool for Producers to Monitor Field-Level Agroclimate for Planning Precision Agriculture Operations. In ASA-CSSA-SSSA annual meeting, November 10-14, 2025. Salt Lake City, UT. (oral presentation).

Vennam, R.R., Raymond, S.[G], Chandel, A.K., Balota, M., Raman, R. 2025. Leveraging Remote Sensing and Machine Learning to Quantify Peanut Leaf Wilting under Heat and Drought Stress. ASA, CSSA, SSSA International Annual Meeting, November 10-14, 2025. Salt Lake City, UT. (oral presentation).

Zheng, Y., Jjagwe, P.[G], Ogando do Granja, M., Chandel, A.K., Ortel, C., Zhang, B., 2025. Integrating UAV-Based Phenotyping and Machine Learning for Enhanced Prediction of Soybean Agronomic Traits. ASA, CSSA, SSSA International Annual Meeting, November 10-14, 2025. Salt Lake City, UT. (poster presentation)

Molaei, B., Lamichhane, B., Gupta, A., Kiraga, S., Chandel, A.K., Peters, T., Khot, L.R., Stockle, C., 2025. Advancing Site-Specific Irrigation with High-Resolution Evapotranspiration Mapping in Washington and Tennessee: AI-Driven Approaches and Ongoing Efforts. ASA, CSSA, SSSA International Annual Meeting, November 10-14, 2025. Salt Lake City, UT. (oral presentation).

Lamichhane, B., Molaei, B., Marek, G., Colaizzi, P., Evett, S., Koff, J., Peters, T., Chandel, A.K., Khot, L.R., Stockle, C., 2025. Predicting the Radiometric Temperature of a Fully Irrigated Canopy Based on Meteorological Data for Improving UAS-Based Evapotranspiration Estimation. ASA, CSSA, SSSA International Annual Meeting, November 10-14, 2025. Salt Lake City, UT. (oral presentation).

Gupta, A., Molaei, B., Marek, G., Colaizzi, P., Evett, S., Peters, T., Chandel, A.K., Khot, L.R., Stockle, C., 2025. Multistate Evaluation of AI-Driven

Radiometric Temperature Prediction of Dry Bare Agricultural Soil Using Meteorological Data for Enhanced UAS-Based Evapotranspiration Mapping. ASA, CSSA, SSSA International Annual Meeting, November 10-14, 2025. Salt Lake City, UT. (oral presentation).

Raman, R., Balota, M., Ernest, E., Jjagwe, P.[G], Vennam, R.R.[G], Chandel, A., 2025. Assessing Faba Bean Emergence Rate, Winter Survival, and Seed Productivity in the Mid-Atlantic. ASA, CSSA, SSSA International Annual Meeting, November 10-14, 2025. Salt Lake City, UT. (poster presentation).

Raman, R., Balota, M., Vennam, R.R.[G], Jjagwe, P.[G], Chandel, A., 2025. Effect of nitrogen treatments on faba bean biomass and grain yield in the mid-Atlantic. ASA, CSSA, SSSA International Annual Meeting, November 10-14, 2025. Salt Lake City, UT. (poster presentation).

Nkwocha, C.[G], Chandel, A.K., Balota, M., Bryant, T., Malone, S., 2025. Identification of Southern Corn Root Worm Injury in Peanuts using Deep convolutional neural network based-YOLO. American Peanut Research and Education Society Meeting, July 15-17, 2025, Richmond, VA. (poster presentation).

Raymond, S.[G], Chandel, A.K., Balota, M., 2025. Precision Peanut Maturity Mapping for Virginia-Type Cultivars using Aerial Spectral Imagery, Weather Data and Advanced Machine Learning. American Peanut Research and Education Society Meeting, July 15-17, 2025, Richmond, VA. (oral presentation).

Jjagwe, P.[G], Chandel, A.K., Balota, M., Raman, R., 2025. Faba bean crop plant identification using aerial multispectral imagery and convolutional neural network-based computer vision models. SPIE Defense + Commercial Sensing exhibition, April 13-17, 2025, Orlando, FL. (oral presentation).

Nkwocha, C.[G], Chandel, A.K., 2025. Unoccupied Ground Vehicle Platform for Plant Disease Risk and Severity Mapping. SPIE Defense + Commercial Sensing exhibition, April 13-17, 2025, Orlando, FL. (oral presentation).

Raymond, S.[G], Chandel, A.K., Balota, M., Chappell, M., Shridhar, V., 2025. Leveraging Stacked Generalization for Peanut Maturity Mapping Using Aerial Multispectral Imagery and Growing Degree Days. SPIE Defense + Commercial Sensing exhibition, April 13-17, 2025, Orlando, FL. (oral presentation).

Jjagwe, P.[G], Chandel, A.K., Seetharaman, R., Balota, M., 2025. Weed identification in Faba bean crop using aerial multispectral imagery and computer vision. AI in Agriculture and Natural Resources Conference, March 31- April 2, 2025, Starkville, MS. (oral presentation).

Nkwocha, C.[G], Chandel, A.K., Bryant, T., Balota, M., 2025. Identification of Southern Corn Root Worm Injury in Peanuts using Deep convolutional neural network based- YOLO. AI in Agriculture and Natural Resources Conference, March 31- April 2, 2025, Starkville, MS. (oral presentation).

Raymond, S.[G], Chandel, A.K., Balota, M., 2025. Advancing Non-Invasive Peanut Maturity Prediction using Aerial Multispectral Imagery and Weather data with stacked ensemble Multi-View Learning. AI in Agriculture and Natural Resources Conference, March 31- April 2, 2025, Starkville, MS. (oral presentation).

Gupta, A., Lamichhane, B., Molaei, B., Chandel, A., Peters, T., Khot, Lav., Stockle, C., 2025. AI-Driven Prediction of Radiometric Temperatures of Dry Bare Agricultural Soil Using Meteorological Parameters. AI in Agriculture and

Natural Resources Conference, March 31- April 2, 2025, Starkville, MS. (poster presentation).

Lamichhane, B., Gupta, A., Molaei, B., Chandel, A., Peters, T., Khot, Lav., Stockle, C., 2025. Methodological Framework for Enhancing UAS-METRIC Evapotranspiration Accuracy Using Machine Learning-Based Anchor Pixel Temperature Estimation. AI in Agriculture and Natural Resources Conference, March 31- April 2, 2025, Starkville, MS. (poster presentation).

Tate, C., Frame, W., Chandel, A.K., Eick, M., Guo, F., Shortridge, J., Stewart, R., 2024. Nitrogen Prediction Algorithm for Cotton Grown in the Upper Southeast Coastal Plain. ASA, CSSA, SSSA International Annual Meeting, Nov 10-13, 2024, San Antonio, TX. (oral presentation).

Chandel, A.K., 2024. Open-source digital platforms towards supporting cropping decisions of producers. 2024 National Agricultural Producer Data Cooperative Conference, August 8-9, 2024, Lincoln, NE. (oral presentation).

Kanakamedala, V.C., Chandel, A.K., 2024. Advancing agriculture communication network in Virginia through a secure web platform "Ag Corp". National Agricultural Producer Data Cooperative Conference, August 8-9, 2024, Lincoln, NE. (poster presentation).

Sarr, A.[G], Chandel, A.K.**,** 2024. Internet-of-Things integrated platform for farm-level agro-climate monitoring and data management. National Agricultural Producer Data Cooperative Conference, August 8-9, 2024, Lincoln, NE. (poster presentation).

Emmanuel, S.R.[G], Jjagwe, P.[G], Balota, M., Chappell, M., Chandel, A.K., 2024. Non-invasive peanut maturity mapping using aerial spectral imaging and artificial intelligence techniques. American Peanut Research and Education Society Meeting, July 8-11, 2024, Oklahoma City, OK. (poster presentation).

Chapu, I., Okello, R.C.O., Okello, D.K., Chandel, A.K.**,** Balota, M., 2024. Application of machine learning for selection for LLS resistance in peanut (Arachis hypogaea L.). American Peanut Research and Education Society Meeting, July 8-11, 2024, Oklahoma City, OK. (poster presentation).

Jjagwe, P.[G], Vennam, R.R., Raymond, S.R.[G], Beard, K., Balota, M., Haak, D., Chandel, A.K., 2024. Smartphone-based thermal-RGB imaging tool to quantify water stress in peanuts. American Peanut Research and Education Society Meeting, July 8-11, 2024, Oklahoma City, OK. (poster presentation).

Vennam, R.R., Chandel, A.K.**,** Balota, M., Beard, K., Haak, D., 2024. Exploring the Feasibility of High Throughput Phenotyping Technology to Enhance Peanut Physiological Resilience for Heat and Drought Tolerance. American Peanut Research and Education Society Meeting, July 8-11, 2024, Oklahoma City, OK. (poster presentation).

Vennam, R.R., Chandel, A.K.**,** Beard, K., Balota, M., 2024. Leveraging UAV Remote Sensing to Enhance Phenotyping of Peanut Physiology for Heat and Drought Tolerance. 24th Annual GIS and Remote Sensing Research Symposium, April 5, 2024. Blacksburg, VA. (poster presentation).

Jjagwe, P.[G], Chandel, A.K.**,** Langston, D., 2024. Quantifying Impact of Nematodes on Soybean Production using Aerial Multispectral Imagery and Machine Learning. 24th Annual GIS and Remote Sensing Research Symposium, April 5, 2024. Blacksburg, VA. (poster presentation).

Emmanuel, S.R.[G]**,** Chandel, A.K.**,** Langston, D., 2024. Corn grain yield mapping using aerial-spectral imagery and machine learning techniques. 24th Annual GIS and Remote Sensing Research Symposium, April 5, 2024. Blacksburg, VA. (poster presentation).

Chandel, A.K., Balota, M., 2023. Feasibility of aerial spectral imagery with ai-machine learning for peanut maturity prediction. ASA, CSSA, SSSA International Annual Meeting, October 29-November 1, 2023, Saint Louis, MO. (oral presentation).

Balota M., Rathore, J., Chandel, A.K., 2023. Does Prohexadione Calcium affect yield, kernel size, and peanut Maturity?. American Peanut Research and Education Society Meeting, July 11-13, 2023, Savannah, GA. (oral presentation).

Chandel, A.K., Balota, M., Rathore, J., 2023. Small UAS based spectral imaging and machine learning for peanut maturity estimation. American Peanut Research and Education Society Meeting, July 11-13, 2023, Savannah, GA. (poster presentation).

Chandel, A.K., 2023. Aerial spectral vegetation indices for cross-crop commodity and cross-crop trait analysis. ASABE Annual International Meeting, July 9-12, 2023, Omaha, NE. (oral presentation).

Chandel A.K., Balota M., 2023. Application of small UAS towards management of field cropping systems of Virginia. USDA S1069: UAV Multistate Annual Meeting, June 1-2, 2023, Suffolk, VA. (oral presentation).

Chandel, A.K., 2023. Building synergistic data framework for cropping, livestock, and aquaculture producers of Virginia. National Agricultural Producers Data Cooperative Conference, May 10-12, 2023, Lincoln, NE. Website and Youtube: University of Nebraska. Retrieved from https://www.agdatacoop.org/conference. (oral presentation).

Chandel, A.K., Mitra, S., Mehul, B., Posadas, B., Chappell, M., Parraga, K., White, R., 2023. Farmers' user experience related to digital advancements in agriculture. National Agricultural Producers Data Cooperative Conference, May 10-12, 2023, Lincoln, NE. (poster presentation).

Chandel, A.K., Balota, M., Rathore, J., 2023. AI-Machine learning with aerial spectral imagery for peanut maturity prediction. AI in Agriculture Conference, April 17-19, 2023, Orlando, FL. (oral presentation).

Deb, S., Chandel, A.K., Langston, D., 2023. Quantification of soybean Cyst Nematode damages using aerial multispectral imaging technique. AI in Agriculture Conference, April 17-19, 2023, Orlando, FL. (poster presentation).

Rathore, J., Chandel, A.K.**,** Langston, D., 2023. Quantifying Sclerotinia blight severity and effective fungicide application strategies in peanuts using aerial multispectral imaging technique. AI in Agriculture Conference, April 17-19, 2023, Orlando, FL. (poster presentation).

Kumari, S., Chandel, A.K.**,** Langston, D., 2023. Evaluation of fungicides and their application effects on corn leafspot using aerial multispectral imaging. AI in Agriculture Conference, April 17-19, 2023, Orlando, FL. (poster presentation).

Mitra, S., Mehul, B., Posadas, B., Chandel, A.K.**,** Chappell, M., Parraga, K., White, R., 2023. Farmers' user experience related to digital advancements in agriculture. AI in Agriculture Conference, April 17-19, 2023, Orlando, FL. (poster presentation).

Kumari S., Chandel, A.K., Langston, D., 2023. Evaluation of fungicides and their application effects on corn leafspot using aerial multispectral imaging. CAIA Big Event, March 16, 2023, Blacksburg, VA. (poster presentation).

Rathore J., Chandel, A.K., Langston, D., 2023. Quantifying Sclerotinia blight severity and effective fungicide application strategies in peanuts using aerial multispectral imaging technique. CAIA Big Event, March 16, 2023, Blacksburg, VA. (poster presentation).

Deb, S., Chandel, A.K., Langston, D., 2023. Quantification of soybean Cyst Nematode damages using aerial multispectral imaging technique. CAIA Big Event, March 16, 2023, Blacksburg, VA. (poster presentation).

Mitra, S., Mehul, B., Posadas, B., Chandel, A.K., Chappell, M., Parraga, K., White, R., 2023. Farmers' user experience related to digital advancements in agriculture. CAIA Big Event, March 16, 2023, Blacksburg, VA. (poster presentation).

Oakes, J., White, R., Chandel, A.K., 2023. Engaging CAIA with VCE. Virginia Cooperative Extension Winter Conference, February 6 – 9, 2023, Blacksburg, VA. (oral presentation).

Tolomelli, G., Kothawade, G.S., Chandel, A.K., Manfrini, L., Jacoby, P., Khot, L.R., 2022, November. Aerial-RGB imagery-based 3D canopy reconstruction and mapping of grapevines for precision management. IEEE Workshop on Metrology for Agriculture and Forestry, November 3-5, 2022, Perugia, Italy. (oral presentation).

Chandel, A.K., Khot, L.R., Blanco, V., Kalcsits, L., Kilham, N.E., Mantle, S., 2022. Multi-scale remote sensing data driven water use and stress mapping of apple trees. Paper No. 2201120, ASABE Annual International Meeting, July 17-20, 2022, Houston, TX. (oral presentation).

Molaei, B., Peters, R.T., Chandel, A.K., Khot, L.R., Stockle, C.O., 2022. Investigating practical artificial hot and cold reference surfaces for improved ET estimation using UAS-METRIC energy balance model. Paper No. 2200629. ASABE Annual International Meeting, July 17-20, 2022, Houston, TX. (oral presentation).

Kothawade, G.S., Chandel, A.K., Khot, L.R., Wright, A.A., Harper, S.J., 2022. Evaluation of high throughput volatile headspace sensing for little cherry disease detection. Paper No. 2200983. ASABE Annual International Meeting, July 17-20, 2022, Houston, TX. (oral presentation).

Chandel, A.K., Langston, D., Emebo, O., 2022. Artificial intelligence & GIS enabled tool for high throughput soybean nematode infestation assessment. CAIA Big Event, March 28, 2022, Blacksburg, VA. (poster presentation).

Chandel, A.K., Khot, L.R., Peters, T.R., Stöckle, C.O., Mantle, S., Kalcsits, L., Blanco, V., Kilham, N.E., 2022. Multi-scale remote sensing and energy balance modeling for geospatial evapotranspiration mapping of apple trees. AI in Agriculture Conference, March 9-11, 2022, Auburn, AL. (poster presentation).

Chandel, A.K., Langston, D., Emebo, O., 2022. Artificial intelligence & GIS enabled tool for high throughput soybean nematode infestation assessment. Envisioning 2050 in the Southeast: AI-Driven Innovations in Agriculture. AI in Agriculture Conference, March 9-11, 2022, Auburn, AL. (poster presentation).

<u>Prior to joining VT</u> (24)

Molaei, B., Peters, R.T., Chandel, A.K., Khot, L.R., Stöckle, C.O., and Campbell, C.S., 2021. Comparing Downwind Evapotranspiration Suppression in LESA and MESA Irrigation Systems. ASABE 6th Decennial National Irrigation Symposium*,* December 6-8, 2021, San Diego, CA. (oral presentation).

Chandel, A.K., Khot, L.R., Brown, D.J., Peters, T.R., Stöckle, C.O., Mantle, S., 2021. Geospatial apple canopy transpiration mapping: effect of in-field and open-field weather. IEEE International Workshop on Metrology for Agriculture and Forestry, November 3–5, 2021, Trento, Italy. Virtual conference. (oral presentation).

Kothawade, G.S., Chandel, A.K., Schrader, M.J., Rathnayake, A., Khot, L.R., 2021. High throughput canopy characterization of a commercial apple orchard using aerial RGB imagery. IEEE International Workshop on Metrology for Agriculture and Forestry, November 3–5, 2021, Trento, Italy. Virtual conference. (oral presentation).

Chandel, A.K., Rathnayake, A., Khot, L.R., 2021. Mapping apple canopy attributes using aerial multispectral imagery for precision crop inputs management. XII International Symposium on Integrating Canopy, Rootstock and Environmental Physiology in Orchard Systems, July 26–30, 2021. Virtual conference. (oral presentation).

Chandel, A.K., Amogi, B.R., Khot, L.R., Peters, R.T., Stöckle C.O., 2021. Cloud compute enabled high resolution ET mapping for grapevine irrigation management. Paper No. 2100683, ASABE Annual International Meeting, July 12-16, 2021. Virtual conference. (oral presentation).

Kothawade, G.S., Chandel, A.K., Schrader, M.J., Rathnayake, A., Khot, L.R., 2021. High throughput canopy characterization of a commercial apple orchard using a small UAS equipped with a consumer-grade camera. Paper No. 2101235, ASABE Annual International Meeting, July 12-16, 2021. Virtual conference. (oral presentation).

Molaei, B., Peters, R.T., Chandel, A.K., Khot, L.R., Stockle, C.O., Campbell, C., 2021. Comparing Downwind Evapotranspiration Suppression in LESA and MESA Irrigation Systems. Paper No. 2100530, ASABE Annual International Meeting, July 12-16, 2021. Virtual conference. (oral presentation).

Chandel, A.K., Khot, L.R., Sallato, B.C. Towards rapid detection and mapping of powdery mildew in apple orchards. IEEE International Workshop on Metrology for Agriculture and Forestry, November 4–6, 2020. Virtual conference. (oral presentation).

Chandel, A.K., Khot, L.R., Stöckle, C.O. Spatiotemporal water use mapping of a commercial apple orchard using UAS based spectral imagery. IEEE International Workshop on Metrology for Agriculture and Forestry, November 4–6, 2020. Virtual conference. (oral presentation).

Amogi. B.R., Chandel, A.K., L.R. Khot, Jacoby, P.W. A mobile thermal-RGB imaging tool for mapping crop water stress of grapevines. IEEE International Workshop on Metrology for Agriculture and Forestry, November 4–6, 2020. Virtual conference. (oral presentation).

Chandel, A.K., Khot, L.R., Molaei, B., Peters, R.T., Stöckle C.O., Jacoby, P.W., Actual evapotranspiration mapping of grapevines using UAS based high resolution imagery and METRIC energy balance model. Paper No. 2001595, ASABE Annual International Meeting, July 12-15, 2020. Virtual conference. (oral presentation).

Molaei, B., Chandel, A.K., Peters, R.T., Khot, L.R., Vargas, J.Q. Crop Lodging Assessment in Irrigated Spearmint Using Low Altitude High-Resolution Imagery. ASABE Annual International Meeting, July 12-15, 2020. Virtual conference. (oral presentation).

Chandel, A.K., Molaei, B., Khot, L.R., Peters, R.T., Stöckle, C.O., High Resolution Geospatial Mapping of Actual Evapotranspiration Using Small UAS Based Imagery for Site Specific Irrigation Management. WaterSmart Innovations Conference and Exposition, October 2-3, 2019*,* Las Vegas, NV*.* Virtual conference. (oral presentation).

Chandel, A.K., Molaei, B., Khot, L.R., Peters, T.R., Stöckle, C.O. Low altitude high resolution remote sensing for mapping actual crop evapotranspiration. AgTech day, August 22, 2019, Prosser, WA. (poster presentation).

Molaei, B., Chandel, A.K., Peters, R.T., Khot, L.R., Khan, A., Maureira, F., Stöckle, C.O., Artificial hot and cold reference surfaces for calibrating crop water use estimation at high spatial resolution. AgTech day, August 22, 2019, Prosser, WA. (poster presentation).

Bahlol, H.Y., Chandel, A.K., Khot, L.R., Hoheisel, G-A. Smart spray analytical system: An efficient tool for air-assisted sprayer calibrations. AgTech day, August 22, 2019, Prosser, WA. (poster presentation).

Chandel, A.K., Molaei, B., Khot, L.R., Peters, R.T., Stöckle, C.O. Low altitude high resolution remote sensing for actual crop evapotranspiration mapping. ASABE Annual International Meeting, July 7-10, 2019*,* Boston, MA*.* (oral presentation).

Molaei, B., Chandel, A.K.**,** Peters, R.T., Khot, L.R., Khan, A., Maureira, F., Stöckle, C.O. Investigation of artificial hot and cold reference surfaces for estimating crop water use at high spatial resolution. ASABE Annual International Meeting, July 7-10, 2019*,* Boston, MA*.* (oral presentation).

Maureira, F., Chandel, A.K., Khan, A., Molaei, B., Khot, L.R., Peters, R.T., Stöckle, C.O. Proximal sensing of crop surface albedo with multispectral imaging sensors on board ground-based platform. ASABE Annual International Meeting, July 7-10, 2019*,* Boston, MA*.* (oral presentation).

Kumar, S.P., Tewari, V.K., Mehta, C.R., Chandel, A.K., Nare, B., Chethan, C.R. Sensor based intra row weeding technology. Asia-Pacific Federation for Information Technology in Agriculture in association with World Congress on Computers in Agriculture, October 24-26, 2019, Indian Institute of Technology Bombay, Mumbai, India. (oral presentation).

Chandel, A.K., Tewari, V.K. Ultrasonic sensor based variable rate pomegranate orchard sprayer. *International conference on emerging technologies in Agricultural and Food Engineering,* December 27-30, 2016, Indian Institute of Technology Kharagpur, India. (oral presentation).

Chandel, A.K., Tewari, V.K., Kumar, S.P., 2015. Digital imaging-based contact type herbicide applicator for row crops. Presented at the *40th Research Scholars Day*, October 2, 2015, Indian Institute of Technology Kharagpur, India. (oral presentation).

Tewari, V.K., Chandel, A.K., Shrivastava, P., 2015. Annual review meeting, *AICRP on FIM* (ICAR, Govt. of India), PAU Ludhiana, India. (oral presentation).

Tewari, V.K., Chandel, A.K., Shrivastava, P., Gupta, C., 2016. Annual review meeting, *AICRP on FIM* (ICAR, Govt. of India), TNAU Coimbatore, India. (oral presentation).

6. Abstracts

Prior to joining VT (1)

Chandel, A.K., Khot, L.R., Molaei, B., Peters, R.T., Stöckle C.O., Drone based Multispectral and Thermal Imagery for Geospatial Water Use Mapping of Irrigated Field Crops. Washington State University Digital Agriculture Summit, October 6–7, 2020, Pullman, WA.

7. Other papers and reports (7 total, 4 since joining VT)

Since joining VT (4)

Clarke, J.[C], Bierman, D., Bucksch, A., Calvert, S., Caprez, A., Chandel, A.K. et al., 2024. National Agricultural Producers Data Cooperative Conference White Paper. https://www.agdatacoop.org/

Samtani, J.[C], Chandel, A.K., Gauthier, N., Rahman, M., 2024. Latent detection of anthracnose on strawberry crop using multi-spectral imaging. 2024 Research Final Report. Southern Region Small Fruit Consortium. https://smallfruits.org/files/2025/01/2024-R-02-final-strawberry-multi-spectral-imaging.pdf

Samtani, J.[C], Chandel, A.K., 2023. Latent detection of anthracnose on strawberry crop using multi-spectral imaging. 2023 Research Final Report. Southern Region Small Fruit Consortium. https://smallfruits.org/files/2023/12/2023-R-15-latent-detection-anthracnose.pdf

Becker, S.[C], Bierman, D., Bucksch, A., Calvert, S., Caprez, A., Chandel, A.K., Craker, B., Dorius, S.F., Elsik, C., Clarke, J., Evans, J., 2023. The NAPDC: Stakeholder Input and Strategic Directions (No. tkg96). Center for Open Science. https://doi.org/10.31219/osf.io/tkg96

Prior to joining VT (3)

Pickering, N.[C], Chandel, A.K., Khot, L.R., Liu, M., Peters, R.T., Kadam, S., Molaei, B., Yoder, J., Rajagopalan, K., Stöckle, C.O., Yorgey, G., Haller, D., Barber, M., Khanal, R., 2021. Evapotranspiration Estimation: Comparison of Evapotranspiration Methods Used in Washington State, The Water Report. Water Rights, Water Quality, & Water Solutions in the West, 208, 10-19. http://www.thewaterreport.com/Issues%20205%20to%20208.html

Tewari, V.K.[C], Chandel, A.K., Shrivastava, P., Gupta, C., 2016. Annual Progress Report, AICRP on FIM (ICAR, Govt. of India).

Tewari, V.K.[C], Chandel, A.K., Shrivastava, P., 2015. Annual Progress Report, AICRP on FIM (ICAR, Govt. of India).

Invited keynote presentations or lectures

Invited keynote lectures (2)

End-to-End Precision Agriculture: Integrating Digital Intelligence, Capacity Building, and Economic Sustainability. In 59th Annual Convention on Engineering Innovations for Agriculture 5.0 and International Symposium on Mechatronics and Robotics in Pre and Postproduction Agriculture. Bhopal, Madhya Pradesh, India. November 11, 2025. (Contact time: 30 min, Attendees: 75).

End-to-End Precision Agriculture Through Digital Intelligence, Capacity Building, and Economic Sustainability. In International Conference on Recent Trends in Smart & Sustainable Agriculture for Food & Nutritional Security. Jalandhar, Punjab, India. November 27, 2025. (Contact time: 45 min, Attendees: 300).

Invited expert guest lectures (In-person/Virtual 33)

Crop Sensing for Sustainable Agricultural Systems. Rsif Guest Webinar Series. Online International Talk. February 25, 2026. (Contact time: 1 h 15 min, Attendees: 75).

End-to-End Precision Agriculture from Farmer Perspective: Integrating Digital Intelligence, Capacity Building, and Economic Sustainability. Dhamtari, Chhattisgarh, India. December 17, 2025. (Contact time: 2 h, Attendees: 200).

Precision Agriculture Discussion with College of Agriculture Undergraduate Students. In Invited Seminar/Workshop. Raipur, Chhattisgarh, India. December 17, 2025. (Contact time: 1 h, Attendees: 150).

End-to-End Precision Agriculture: Integrating Digital Intelligence, Capacity Building, and Economic Sustainability. In Invited Seminar. IIIT Vadodara, Ahmedabad campus, Gujarat, India. December 9, 2025. (Contact time: 1 h, Attendees: 30).

End-to-End Precision Agriculture: Integrating Digital Intelligence, Capacity Building, and Economic Sustainability. Rahuri, Maharashtra, India. December 4, 2025. (Contact time: 1 h, Attendees: 25).

End-to-End Precision Agriculture: A Pathway for Young Scientists in the 21st Century. In International Agronomy Congress. New Delhi, India. November 24-26, 2025. (Contact time: 20 min, Attendees: 50).

The sixth sense of crops: "Sensors" for sustainable agriculture. In A workshop on Indigenous Sensors and Actuators Development for IoT Ecosystem. Online, India. July 30, 2025. (Contact time: 1 h, Attendees: 75).

Precision Ag at Virginia Tech Tidewater AREC for Peanuts. Discussion with peanut stakeholders from Argentina. July 18, 2025. Tidewater AREC. (Contact time: 30 min, Attendees: 5).

CSES 4144 (Plant Breeding and Genetics): Drones in Agriculture. Virginia Tech, Blacksburg. February 19, 2025. (Contact time: 1 h, Attendees: 30).

Free webtool for producers to monitor field-level agroclimate for precision agriculture operations, National Agricultural Producers Data Cooperative (NAPDC) webinar. February 17, 2025. (Contact time: 40 min, Attendees: 30).

Era of precision agriculture technologies based on multimodal sensing and artificial intelligence. November 20, 2024. Indian Institute of Technology Indore, India. (Contact time: 1 h, Attendees: 45).

BSE 5944 (Graduate Seminar): Agricultural Research and Extension Centers (ARECs) & Precision Ag and Data Management program. October 1, 2024. (Contact time: 45 min, Attendees: 20).

User-inspired digital agriculture: prospects and challenges. C-DAC Kolkata, India. November 12, 2024. (Contact time: 1 h, Attendees: 60).

Agriculture 4.0: prospects of data-intensive farming. Indian Institute of Technology Kharagpur, India. November 11, 2024. (Contact time: 1 h 30 min, Attendees: 45).

User-inspired digital agriculture in the context of Indian farmers. Indian Institute of Technology Bhubaneshwar, India. November 9, 2024. (Contact time: 1 h 30 min, Attendees: 45).

User-inspired digital agriculture through multimodal data and artificial intelligence. IEEE Roorkee, sub-section, and IEEE Uttar Pradesh section GRSS chapter. Indian Institute of Technology Roorkee. January 25, 2024. (Contact time: 1 h 30 min, Attendees: 45).

Modern-day production agriculture management using multimodal data and artificial intelligence. Winter School for scientists on recent advances in electronic devices, artificial intelligence, and machine learning for precision agriculture. ICAR-Central Institute of Agricultural Engineering. Bhopal, India. January 21, 2024. (Contact time: 1 h 30 min, Attendees: 45).

The experiences of scientific writing process. Winter School for scientists on recent advances in electronic devices, artificial intelligence, and machine learning for precision agriculture. Organized by ICAR-Central Institute of Agricultural Engineering. Bhopal, India. January 21, 2024. (Contact time: 1 h 30 min, Attendees: 45).

User-inspired digital agriculture using multimodal data and artificial intelligence. Invited theme talk for “AI and Hyperspectral Imaging” technical session. International conclave on futuristic farming. December 20-21, 2023. (Contact time: 30 min, Attendees: 100).

Digital agriculture and artificial intelligence. Indian Institute of Technology Kharagpur, India. December 18, 2023. (Contact time: 1 h 30 min, Attendees: 50).

Evolving world of technology and data-run solutions to support agricultural operations. Virginia Governor's School for Agriculture. July 21, 2023. Blacksburg, VA. (Contact time: 1 h, Attendees: 100).

BSE 5944 (Graduate Seminar): Technologies for next generation Ag. April 10, 2023. (Contact time: 45 min, Attendees: 20).

Drones for precision mapping and spraying: Current Status, Challenges, & Future Roadmaps. National Institute of Plant Health Management (NIPHM), Hyderabad, India. March 25, 2023. (Contact time: 1 h 30 min, Attendees: 120).

The next-gen ag: “digital ag”. Indian Society of Agricultural Engineering (ISAE) Bhopal chapter. March 6, 2023. (Contact time: 1 h 30 min, Attendees: 120).

The fundamentals of scientific writing. Indian Society of Agricultural Engineering (ISAE) Bhopal chapter. March 6, 2023. (Contact time: 1 h 30 min, Attendees: 120).

Center for Advanced Innovation in Agriculture, National Agricultural Producers Data Cooperative (NAPDC) webinar. September 19, 2022. (Contact time: 1 h 30 min, Attendees: 30).

Smart Agriculture Virtual Panel Discussion. MITRE Engenuity Open Generation 5G Summit. September 13, 2022. (Contact time: 1 h 30 min, Attendees: 50).

Sensors and data analytics for agriculture. Governor's School for Agriculture Elective Course [CAIA], Virginia Tech. USA. July 18-20, 2022. (Contact time: 1 h, Attendees: 50).

Sensors and data applications for smart agriculture. C-Tech^2 summer program for rising high school juniors and seniors. June 30, 2022. (Contact time: 25 min, Attendees: 35).

Next Gen Ag to Realize "Farms of the Future". Azadi Ka Amrit Mahotsav Webinar Series, Indian Institute of Technology Kharagpur, India. May 20, 2022. (Contact time: 1 h, Attendees: 55).

Building climate smart tools for agriculture. Prakriti-Agri Innovation Fest, Indian Institute of Technology Kharagpur, India. April 1, 2022. (Contact time: 1 h 30 min, Attendees: 35).

Precision Ag technologies to realize "Farms of the future". Winter School for scientists on recent advances in electronic devices, artificial intelligence, and machine learning for precision agriculture. Organized by ICAR-Central Institute of Agricultural Engineering. Bhopal, India. February 19, 2022. (Contact time: 1 h 30 min, Attendees: 45).

High resolution remote sensing and energy balance modeling for crop evapotranspiration mapping. Winter School for scientists on recent advances in electronic devices, artificial intelligence, and machine learning for precision agriculture. Organized by ICAR-Central Institute of Agricultural Engineering. Bhopal, India. February 19, 2022. (Contact time: 1 h 30 min, Attendees: 45).

<u>Invited expert panel discussions</u> (6)

Agricultural systems 5.0 for sustainable productivity. In 59th Annual Convention on Engineering Innovations for Agriculture 5.0 and International Symposium on Mechatronics and Robotics in Pre and Postproduction Agriculture. Bhopal, Madhya Pradesh, India. November 10, 2025. (Contact time: 30 min, Attendees: 75).

Emergence, application, and future of precision agriculture technologies. Agripalooza. March 18, 2025. Botetourt County, VA.

Precision Ag for enhancing farming productivity and quality of life. Precision Ag Technology Expo. June 30, 2024. Suffolk, VA.

Smart farming technologies. Broadband Together Conference. May 17, 2024. Richmond, VA.

Evolving world of technology and data-run solutions for agriculture. Virginia Governor's School for Agriculture. July 21, 2023. Blacksburg, VA.

Smart Agriculture. MITRE Engenuity Open Generation 5G Summit. September 13, 2022.

C. Editorships, curatorships, etc.

1. Journals or other learned publications

2025-2026: Topic Editor, Frontiers in Plant Science: Advanced Technologies in Precision Agriculture for Optimizing Crop Interactions with Pesticides and Herbicides

2025-2026: Guest Editor, Computers: Intelligent Computing and Sensing Systems for Sustainable Precision Agriculture

2023-2024: Guest Editor, Agriculture: Precision Agriculture Technologies for Crop Management

2022-2024: Associate Guest Editor, Journal of Field Robotics: AI and Ergonomics in Agriculture

2. **Editorial boards**

2025-present: Associate Editor, Frontiers in Plant Science: Technical Advances in Plant Science
2025-present: Associate Editor, Irrigation Science

D. Intellectual properties

1. **Software** (including open-source software)

Agroclimate viewer and planner app (AgroVAP): A free Google Earth Engine–based platform that integrates crop health imagery, soil data, historical weather, and 16-day hourly forecasts into a unified interface. Using research-driven algorithms, AgroVAP delivers actionable outputs to users (farmers, crop consultants, researchers, and other agricultural stakeholders) including, crop growth tracking, disease risk alerts, and irrigation and spray scheduling recommendations for crop input optimization.
https://datl-chandel.github.io/Agroclimate/.

2. **Patents**

Sensor based intra-row weeding system. Patentee: Indian Institute of Technology Kharagpur. Authors: Tewari, V.K.; Kumar, S.P.; Nare, B.; Chandel A.K.; Chethan, C.R. Indian Patent No. 508805; Application No. 201831024844 A.

This patented invention introduces a sensor-based intra-row weeding system that overcomes a critical limitation in precision agriculture of selective weed removal within crop rows without damaging crops. Unlike conventional inter-row systems or static mechanical weeders, this technology integrates real-time plant-detection sensors with a dynamically actuated vertical-axis rotary weeder mounted on a four-bar linkage mechanism. Upon detecting a crop plant, a control unit actuates a PMDC motor to laterally shift the weeder out of the plant's path, automatically returning it to position once the plant passes. This sensing–actuation integration represents a novel advancement over prior art by enabling intelligent, adaptive intra-row weeding. The system reduces herbicide dependence, lowers labor requirements, minimizes crop injury, and enhances operational efficiency, demonstrating strong translational potential for sustainable and automated crop production systems.

3. **Provisional-patent**

A Modular, 3D-Printed Microneedle Biosensing Platform with Lithography-Free Electrodes for Wireless, In-Situ Monitoring of Biotic and Abiotic Stresses in Crops Authors: Chandel A.K.; Erukainure, F[G]. VTIP 26-081.

This invention introduces a wearable, electrochemical biosensing platform for continuous, in-situ plant health monitoring, distinguished by a novel, lithography-free fabrication process that combines Stereolithography (SLA) 3D printing with direct electrodeposition of noble metals to create robust, low-cost electrodes. Unlike traditional sensors that rely on expensive clean-room photolithography or destructive tissue sampling, this device utilizes an integrated microneedle array and microfluidic channel for minimally invasive extraction of vascular sap, enabling real-time detection of both biotic (e.g., fungal pathogens) and abiotic (e.g., water, nitrogen) stresses. The platform's primary innovation is its modular surface chemistry, featuring a universal transducer layer that supports rapid functionalization with interchangeable biorecognition coatings such as enzymes, antibodies, or rGO–chitosan allowing for versatile, multiplexed sensing and wireless integration for data-driven precision agriculture.

IV. International and Professional Service and additional Outreach and Extension Activities

A. International programs accomplishments

1. Other international activities

2025-present Member, International Organization for Standardization US TAG Technical Committee 347 on Data-Driven Agrifood Systems

2025 International Advisory Committee, 59th Annual Convention on Engineering Innovations for Agriculture 5.0 and International Symposium on Mechatronics and Robotics in Pre- and Post-Production Agriculture 2025. Indian Council of Agricultural Research-Central Institute of Agricultural Engineering, Bhopal, India.

2024 International Advisory Committee, IEEE India Geoscience and Remote Sensing Symposium 2024. Goa, India.

B. Professional service accomplishments

1. Service as an officer of an academic or professional association:

<u>Multistate Research Teams</u> (Hatch Project)
2025-present Chair, S1098: Autonomy for Agricultural Production, Processing, and Research to Advance Food Security through Sustainable and Climate-Smart Methods

<u>SPIE, The International Society for Optics and Photonics</u>
2026-present Conference Chair, Autonomous Air and Ground Sensing Systems for Agricultural Optimization and Phenotyping XI.

2. Other service to one's profession or field (e.g., service on committees)

<u>Multistate Research Teams</u> (Hatch Project)
2024-present Member and VA representative, S1090: AI in Agroecosystems: Big Data and Smart Technology-Driven Sustainable Production
2023-present Member, S1069: S1069: Research and Extension for Unmanned Aircraft Systems (UAS) Applications in U.S. Agriculture and Natural Resources
2026-present Chair, Ad Hoc Committee for Drafting Evaluation Template for Future AI in Ag. Conference Host Selection

<u>American Peanut Research and Education Society</u>

2025-present Member, Coyt T. Wilson Distinguished Service Award Committee
2025-present Member, Technical Committee

American Society of Agricultural and Biological Engineers
2025-present Member, ASABE Extension Committee EOPD 208
2025-present Member, Circular Bioeconomy Systems Institute
2021-present Member, MS-60 Unmanned Aerial Systems
2021-present Member, Autonomous Field Equipment
2021-2023 Member, Young Professionals Community

Meeting/Conference organization and service

2024 Session moderator/chair: Sensors and robotics in agriculture. 2024 AI in Agriculture and Natural Resources: Innovation and Discovery to Manage Sustainability in a New World of Environmental Stress. April 15-17. College Station, TX.

2023 Judge: ASABE AIM student short paper competition 2023.

2023 Judge: Student Poster Session, AI in agriculture: innovation and discovery to equitably meet producer needs and perceptions. April 17-19. Orlando, FL.

2022 Judge: AI Driven Tools and Technologies for High Throughput Phenotyping, Oral Session, ASABE Annual International Meeting. July 17-20. Houston, TX.

2022 Judge/Reviewer: ITSC-Information Technology, Sensors & Control Systems, Oral Session, ASABE Annual International Meeting. July 17-20. Houston, TX.

2022 Judge/Reviewer: ITSC-Information Technology, Sensors & Control Systems, Poster Session, ASABE Annual International Meeting. July 17-20. Houston, TX.

2021 Moderator, ITSC-Information Technology, Sensors & Control Systems Virtual Poster-B session. ASABE Annual International Meeting.

2019 Co-moderator, 316 Precision Irrigation, ASABE Annual International Meeting. July 7-10. Boston, MA, USA.

3. Professional meetings, panels, workshops, etc., led or organized

Session and Panel Chair, Autonomous Air and Ground Sensing Systems for Agricultural Optimization and Phenotyping XI. SPIE DCS Conference 2026, National Harbor, MD. April 26-30, 2026. https://spie.org/DS26/conferencedetails/autonomous-air-and-ground-sensing-systems-for-agricultural-optimization-and-phenotyping

Technical Organizing Committee, AI in Agriculture 2026 Conference. Raleigh, NC. March 31-April 2, 2026. https://units.cals.ncsu.edu/2026-ai-ag-conference/

C. Efforts to expand the discipline such as:

D. Additional outreach and extension activities and outcomes

1. Professional achievements in program development, implementation, and evidence of impact

AgroVAP: Agroclimate Viewer and Planner App

Program development
To address persistent cost, accessibility, and technical barriers limiting adoption of digital decision-support tools in agriculture, I conceived and led the development of AgroVAP (Agroclimate Viewer and Planner App), a free, globally accessible web-based decision-support platform (https://datl-chandel.github.io/Agroclimate/). AgroVAP integrates multi-source datasets including farm-level satellite imagery, soil properties, historical and real-time weather observations, and 16-day hourly forecasts into a unified analytical environment designed to support precision crop management. This initiative represents an original and translational contribution that operationalizes my research in digital agriculture, agroclimate analytics, and data-driven crop systems into a publicly available tool. By intentionally eliminating subscription costs and technical barriers, AgroVAP democratizes access to advanced agroclimate intelligence, particularly for small- and medium-scale producers who are often excluded from proprietary precision agriculture platforms. The program reflects an integrated Research–Extension model that bridges discovery, digital innovation, and stakeholder-centered application.

Implementation
I directed the complete lifecycle of AgroVAP, including conceptual design, system architecture, data integration, validation, public deployment, and stakeholder dissemination. Since its launch in February 2025, the platform has reached approximately 980 unique users and generated more than 2,300 visits as of March 6, 2026. User engagement metrics indicate sustained and meaningful utilization, with users returning an average of 2.3 times and spending approximately 19 minutes per session demonstrating substantive interaction and integration into operational decision workflows. Strategic dissemination has included live demonstrations, hands-on workshops, and Extension programming to facilitate stakeholder adoption. Post-demonstration evaluation surveys indicate that participants consistently report they have not encountered a decision-support tool that is simultaneously accessible, integrated, and user-friendly. With all live demonstrations and workshops delivered to date, on a scale of 1 to 5, over 80% of attendees reported high gains in knowledge and a strong likelihood (average rating of 4) of adopting the platform in their management practices. The platform's web-based architecture ensures global accessibility, positioning the program as both a state-serving resource and a scalable digital infrastructure with national and international applicability.

Evidence of impact and significance
AgroVAP demonstrates clear and quantifiable evidence of adoption, behavioral change, and economic relevance. While demonstration and workshop evaluations reflect immediate interest and high perceived value, follow-up web-based user surveys indicate sustained impact: approximately 50% of respondents

report actively integrating the platform into on-farm management decisions, and an additional 30% express near-term intent to adopt, signaling strong adoption momentum and continued demand. Platform analytics further validate substantive utilization, as repeat visitation patterns and extended session durations indicate that stakeholders are transitioning from exploratory use to sustained reliance on AgroVAP as an operational decision-support tool rather than a static informational resource.

By enabling data-driven decisions in disease forecasting, irrigation timing, and precision input optimization, AgroVAP has the modeled potential to reduce production inputs including labor, water, chemicals, and energy by up to 30%, while increasing net farm returns by approximately 20% (estimated at ~$200 million annually in Virginia alone). These projections demonstrate substantial economic and environmental benefits at the state level. Importantly, because AgroVAP is globally accessible and freely available, its potential impact extends beyond regional boundaries, supporting producers and agricultural professionals worldwide. Collectively, this program exemplifies successful translation of research into scalable digital infrastructure, establishes leadership in climate-smart and precision agriculture innovation, and delivers quantifiable economic, educational, and sustainability outcomes. AgroVAP represents a distinctive and forward-looking scholarly contribution that advances digital transformation in agriculture at state, national, and global scales.

2. **Outreach and extension publications, including trade journals, newsletters, websites, journals, multimedia items, etc.** (28 total, 22 since joining VT, Lead author bolded; advisees under my direct mentorship are indicated – [P]postdoc advisee, [G]graduate student advisee; [C]corresponding author)

Trade journals, newsletters, and magazine publications

Since joining VT (7)

Chandel, A.[C] 2025. Agroclimate viewer and planner app. Tidewater AREC newsletter. National Agricultural Producers Data Cooperative newsletter. June 2025. https://static1.squarespace.com/static/61841227e123d82aee09429e/t/69383eb57f3dcd591d836a2d/1765293749550/NAPDC_JuneNewsletter.pdf

Chandel, A.[C] 2025. Agroclimate viewer and planner app. Virginia Tech Tidewater AREC newsletter. https://www.arec.vaes.vt.edu/content/dam/arec_vaes_vt_edu/tidewater/2025-tarec-newsletters/March%2010%202025%20TAREC%20Newsletter.pdf

Aljawasim, B., Richardson, P., Samtani, J.[C], Chandel, A.K., 2024. Researchers see promise in multi-spectral imaging for latent detection of anthracnose disease on strawberry crop. Virginia Strawberry Association news, 9(1), pp.12-13.

Aljawasim, B., Richardson, P., Samtani, J.[C], Chandel, A.K., 2024. Detection of anthracnose disease in strawberries. Southern Region Small Fruit Consortium. https://smallfruits.org/2024/01/detection-of-anthracnose-disease-in-strawberries/

Chandel, A.K.[C], Balota M., Jjagwe, P.[G], Chappell, M., 2024. Pod maturity assessments for Virginia type peanuts: 2023 season results. Virginia-

Carolinas peanut news. 72(1), p.14. https://www.aboutpeanuts.com/peanut-newsletter/166-2024-winter-peanut-news-vol-72-no-1

Chandel, A.K.[C], Amogi, B., Khot, L.R, Stöckle, C.O., Peters, R.T., 2022. Digitizing Crop Water Use with Data-Driven Approaches. ASABE Resource, July/Aug 2022, 269, 14–16. https://elibrary.asabe.org/abstract.asp?aid=53518

Chandel, A.K.[C], Khot, L.R., Kilham, N., Kalcsits, L., Mantle, S., Peters, R.T., Stöckle, C.O., 2022. Multi-scale remote sensing data driven apple crop water use mapping. Washington State University – Fruit Matters, June 2022. https://treefruit.wsu.edu/?post_type=article&p=30922

Prior to joining VT (3)

Khot, L.R.[C], Chandel, A.K., Peters, T.R., Stöckle, C.O., 2021. Drone-based Grapevine Water Use Mapping. Washington State University - Viticulture and Enology Extension News, Spring 2021. http://s3-us-west-2.amazonaws.com/sites.cahnrs.wsu.edu/wp-content/uploads/sites/66/2021/04/15161804/2021-Spring-VEEN-FINAL.pdf

Chandel, A.K.[C], Khot, L.R., Peters, T.R., Stöckle, C.O., Mantle, S., 2021. High spatiotemporal apple crop water use mapping using drone imagery. Washington State University – Fruit Matters, August 2021. https://treefruit.wsu.edu/?post_type=article&p=26820

Tewari, V.K.[C], Nare, B., Kumar, S.P., Chandel, A.K., Tyagi, A., 2014. A six-row tractor mounted microprocessor-based herbicide applicator for weed control in row crops. International Pest Control, 56(3), 162–166. https://search.proquest.com/openview/15d33dbc8a6561e3523abbbd059560ab/1?pq-origsite=gscholar&cbl=2029999

Multimedia releases and mentions

Since joining VT (15)

"New agroclimate app delivers real-time climate and crop insights" by Suzanne M. Pruitt. Virginia Tech News. February 25, 2026. https://news.vt.edu/articles/2024/09/aminata-sarr-collaboration.html

AgroVAP: Free Webtool for Producers. Focus on Cotton Webcast by APS Grow Plant Health program. November 25, 2025. https://doi.org/10.1094/GROW-COT-11-25-502

"AgroVAP" what it means for agricultural producers. Eastern Virginia Crop Talk Podcast. September 29, 2025. https://open.spotify.com/episode/4kX2o4fRmsyHhK99RCa0wP

Agroclimate viewer and planner app. Youtube. February 11, 2025. https://www.youtube.com/watch?v=nZ7mCx14ME4&ab_channel=DigitalAgricultureTechnologiesLab

Agroclimate viewer and planner app (NDVI Only). Youtube. February 11, 2025. https://www.youtube.com/watch?v=pB3WCnZKYj8&ab_channel=DigitalAgricultureTechnologiesLab

"Aminata Sarr cultivates precision agriculture innovations at the Tidewater Agricultural Research and Extension Center" by Suzanne M. Pruitt. Virginia Tech News. September 24, 2024. https://news.vt.edu/articles/2024/09/aminata-sarr-collaboration.html

"Tidewater Agricultural Research and Extension Center hosts international researchers for Drone School program" by Suzanne M. Pruitt. Virginia Tech

News. June 28, 2024. https://news.vt.edu/articles/2024/06/tidewater-arec-drone-school-program.html

"Community Conversation-Tidewater Ag & Research Extension Center". June 19, 2024. Real Country 101.7. https://bit.ly/AgExpoConversation.

"VT Tidewater AREC hosts drone school for six international visitors". Suffolk Herald. April 18, 2014. https://www.suffolknewsherald.com/2024/04/18/vt-tidewater-arec-hosts-drone-school-for-six-international-visitors/

"Virginia Tech Agricultural Research and Extension Centers to host annual field days" by Marya Barlow. Virginia Tech News. March 19, 2024. https://news.vt.edu/articles/2024/03/cals-AREC-field-days.html

"Researchers awarded $2.7 million grant to develop the faba bean as a sustainable mid-Atlantic crop" by Marya Barlow. Virginia Tech News. January 15, 2024. https://news.vt.edu/articles/2024/01/cals-faba-bean-grant.html

"Building synergistic data framework for cropping, livestock, and aquaculture producers of Virginia", NAPDC annual conference and board meeting. May 11, 2023. https://www.youtube.com/watch?v=v256o3nfUQo

"Virginia Tech researchers use artificial intelligence to combat soybean nematodes" By Max Esterhuizen (photos by Jon Eisenback). Virginia Tech News. October 24, 2022. https://vtx.vt.edu/articles/2022/10/cals-nematodes.html

"Smart Agriculture Panel Discussion", 2022 Open Generation 5G Summit. September 27, 2022. https://www.youtube.com/watch?v=DrSkWRMY_do&t=312s

"SmartFarm Innovation Network sees significant increase in talented researchers" by By Mary Hardbarger. Virginia Tech News. September 12, 2022. https://news.vt.edu/articles/2022/08/cals-smartfarm-hires-2022.html

Prior to joining VT (3)

"Sugarcane Planter" by Super Kisan Youtube Channel. January 11, 2021. https://www.youtube.com/watch?v=VEC53GC7jAg&ab_channel=SUPERKISAN

"Smart device developed for precise use of herbicides" by EMN, Science and Tech, Eastern Mirror. May 3, 2018. https://easternmirrornagaland.com/smart-device-developed-for-precise-use-of-herbicides/

"Smart device developed for precise use of herbicides" by Manu Moudgil, Down To Earth. May 2, 2018. https://www.downtoearth.org.in/news/agriculture/smart-device-developed-for-precise-use-of-herbicides-60392

3. Presentations in area of expertise to community and civic organizations, including schools and alumni groups, etc.

Chandel, A.K. Precision agriculture for sustainability. AgTech Summer Camp, Virginia Tech Ronoake Center. June 8-10, 2026. Ronoake VA. (Contact time: ~2.5 h, Attendees: ~20).

Chandel, A.K. Drones for agriculture. Kings Fork high school class lecture. January 21, 2026. Suffolk VA. (Contact time: ~1.5 h, Attendees: ~20).

V. University Service

A. Department, college, and university service, including administrative responsibilities

University
2022-2026 Presidential Post-doctoral Fellowship Review Committee.

College
2023-present Member, CAIA extension and producer engagement committee.

Department

Biological Systems Engineering
2022-Present BSE Honorifics Committee
2022-Present BSE Extension Committee

Tidewater Agricultural Research and Extension Center (AREC)
2024 Search committee member, Agricultural Specialist III, Crop Physiology Lab
2023 Search committee member, Extension Entomology Specialist
2023 Search committee member, post-doctoral associate, Crop Physiology Lab